

FIBER-GRAND: Exact Shell Certification for Aggregate-ML Decoding with One Deletion and Substitutions

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Abstract—Exact likelihoods for insertion/deletion/substitution channels are obtained by summing over compatible synchronization paths, so the first codeword reached by a single likely history need not be maximum likelihood (ML). We isolate the finite-block model of exactly one uniformly located deletion followed by independent binary symmetric substitutions and an arbitrary binary code accessed through membership tests. We give an explicit two-codeword family in which the largest deletion-alignment component and the aggregate-ML codeword strictly reverse, with threshold $np_s > 1 + 2p_s^2$. We then specialize inverse-noise search to aggregate candidate likelihoods. Its channel-specific ingredients are a closed-form upper bound on every word unseen after a completed substitution shell, an exact complete-tie stopping certificate, and a code-independent realization-dependent stopping bound. Standard Hamming-volume asymptotics then give the usual $h_2(p_s)$ high-probability query exponent as a corollary, not as a new entropy law. A random-code occupancy calculation predicts the observed number of completely scored codewords. Across 996 exact finite-length trials, all exact comparators agreed; at $n = 32$ the median number of completely scored codewords was 1.5–13.5. Operation counts and wall-clock time are reported separately, and no universal latency or code-specific optimality claim is made.

I. INTRODUCTION

For variable-length channels, the likelihood of an output is obtained by marginalizing over compatible edit paths; this principle and efficient synchronization-channel likelihood computation are classical [1], [2]. In the pure-deletion limit, ML decoding is governed by alignment or embedding multiplicity [4]. This differs from GRAND on an additive channel, where likelihood-ordered noise guessing makes the first codebook hit ML [3]. Thus, GRAND’s first-hit argument does not transfer unchanged when multiple synchronization histories coalesce into one candidate word.

Related work includes ML formulations and certified optimization for binary codes with deletion synchronization errors [5], sequential synchronization decoding [7], and a shift-compensating minimum distance over one-deletion alignments that is linearly computable [6]. Most-probable-path versus most-probable-string distinctions and incumbent-versus-unfinished-mass bounds are also

established in probabilistic automata [8]. Generic threshold stopping is classical [9]; recent guessing-codeword work likewise proves ML under appropriate incumbent bounds [10]. We therefore do not claim a new aggregate-likelihood principle, a new deletion distance, or a new generic stopping paradigm.

The narrow contributions are as follows. First, we give a strict binary one-deletion-plus-BSC construction with an exact reversal threshold. Second, for this channel we derive a closed-form unseen-word bound and a complete-tie exact certificate for membership-based inverse search. Third, we prove a realization-dependent stopping theorem and finite-confidence work caps. The associated $h_2(p_s)$ exponent is explicitly presented as standard Hamming-volume asymptotics applied after that stopping result. Fourth, we provide exact finite-length evidence and a random-code occupancy benchmark that explains the observed scoring work. The model is fixed to one uniform deletion and hard BSC substitutions; multiple edits, soft information, code design, and hardware are outside scope.

II. CHANNEL LAW AND A STRICT ALIGNMENT-COMPONENT REVERSAL

Let $X = (X_1, \dots, X_n) \in \{0, 1\}^n$. Independent substitutions with probability $0 \leq p_s < 1/2$ act before exactly one uniformly located deletion. The observation $Y \in \{0, 1\}^m$ has $m = n - 1$. The nonempty code $\mathcal{C} \subseteq \{0, 1\}^n$ has a uniform prior and is accessed through $\mathbf{1}\{x \in \mathcal{C}\}$. For candidate x and observation y , define

$$d_j(x, y) = \sum_{i < j} \mathbf{1}\{x_i \neq y_i\} + \sum_{i > j} \mathbf{1}\{x_i \neq y_{i-1}\}. \quad (1)$$

Conditioning on the deletion coordinate gives the established aggregate form specialized to this channel,

$$W(y \mid x) = \frac{1}{n} \sum_{j=1}^n p_s^{d_j(x, y)} (1 - p_s)^{m - d_j(x, y)}. \quad (2)$$

All d_j are obtained in $O(n)$ arithmetic operations from

$$d_{j+1} = d_j - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}. \quad (3)$$

The quantity below is the largest *deletion-alignment component* after the unobserved substitution on the deleted coordinate is marginalized. A fully specified pre-deletion

history adds the same maximizing factor $1 - p_s$ to every candidate and does not change the ordering.

Theorem 1 (Strict component/aggregate-ML reversal). *For $n \geq 4$ and $0 < p_s < 1/2$, let*

$$y = 0^{n-3}10, \quad x_A = 0^{n-3}101, \quad x_B = 0^n, \quad \mathcal{C} = \{x_A, x_B\}.$$

The largest deletion-alignment component strictly favors x_A , while aggregate ML favors x_B if and only if

$$np_s > 1 + 2p_s^2. \quad (4)$$

Proof sketch. Deleting the final symbol of x_A requires no survivor substitution, while every alignment of x_B requires one. The mismatch multiplicities of x_A at distances 0, 1, 2, 3 are 1, 1, 1, $n - 3$. Substitution in (2) gives

$$\frac{n\{W(y | x_B) - W(y | x_A)\}}{(1 - p_s)^{n-4}} = (1 - 2p_s)(np_s - 1 - 2p_s^2),$$

which proves the statement. Complete proofs are in the supplement. $\square$

At the experimental point $p_s = 0.05$, (4) holds exactly for integer $n \geq 21$; therefore this family reverses at $n = 24, 32$ but not at $n = 16$. The general path-versus-aggregate phenomenon is known; the claim here is only the explicit strict binary family and threshold.

III. SHELL-CERTIFIED EXACT AGGREGATE SEARCH

For $z \in \{0, 1\}^m$, let $\mathcal{I}(z)$ be its binary one-insertion sphere. Its classical cardinality is $n + 1$ [11]. Define

$$h_y(x) = \min_j d_j(x, y). \quad (5)$$

Up to orientation, this is the known minimum one-deletion alignment distance used in [6]. The immediate inverse-enumeration property needed here is: if shell s enumerates every error pattern e of weight s , forms $z = y \oplus e$, and generates $\mathcal{I}(z)$, then candidate x first appears at shell $h_y(x)$.

For $0 < p_s < 1/2$, after completing shell $s < m$, every unseen candidate has $d_j \geq s + 1$ for all j . Hence its aggregate likelihood obeys the code-independent bound

$$W(y | x_{\text{unseen}}) \leq U_s := p_s^{s+1}(1 - p_s)^{m-s-1}. \quad (6)$$

The factor $1/n$ in (2) cancels the sum over n alignments, so no extra factor n appears in U_s .

Theorem 2 (Exact unseen-shell certificate). *After shell $s < m$, every unseen word has likelihood at most U_s . If a completely scored incumbent has likelihood strictly greater than U_s , the incumbent set is the complete global ML tie set. Algorithm 1 always terminates and returns exact aggregate ML.*

Proof. First appearance at $h_y(x)$ implies $d_j \geq s + 1$ for every alignment of an unseen word. Since $p_s/(1 - p_s) < 1$, every term in (2) is at most U_s , and so is their average. Strict comparison excludes both a better unseen codeword

Algorithm 1 Shell-certified one-deletion FIBER-GRAND

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1: seen  $\leftarrow \emptyset$ ; no incumbent
2: for  $s = 0, 1, \dots, m$  do
3:   for all  $e \in \{0, 1\}^m$  with  $\text{wt}(e) = s$  do
4:      $z \leftarrow y \oplus e$ 
5:     for all  $x \in \mathcal{I}(z)$  using the  $n + 1$  distinct
       generator do
6:       if  $x \notin \text{seen}$  then
7:         insert  $x$  into seen and query  $x \in \mathcal{C}$ 
8:       if  $x \in \mathcal{C}$  then
9:         evaluate the complete score (2) using
10:      (3)
11:      update the incumbent score and com-
       plete tie set
12:   if an incumbent exists and  $[s = m \text{ or } \text{incumbent}$ 
        $> U_s]$  then
13:     return the incumbent and complete ML tie set

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and an unseen tie. At $s = m$, every binary length- n word has been generated. $\square$

Strictness is necessary for a complete tie set. For example, at $n = 4$, $y = 001$, $p_s = 1/3$, and $\mathcal{C} = \{0010, 0000\}$, the first word is seen at shell zero, the second is unseen, and both have likelihood $U_0 = 4/27$. A non-strict certificate would omit an ML tie. At $p_s = 0$, shell zero contains every positive-likelihood word; at $p_s = 1/2$, strict shell decay disappears and complete tie certification requires exhaustion.

IV. STOPPING, FINITE WORK, AND THE STANDARD ENTROPY COROLLARY

Let T be the realized number of substitutions among the m surviving symbols, let $\rho = p_s/(1 - p_s)$, and define

$$L_n(p_s) = \min\{\ell \geq 1 : \rho^\ell < 1/n\} = \left\lceil \log_{(1-p_s)/p_s} n \right\rceil + 1. \quad (7)$$

Theorem 3 (Realization-dependent stopping). *For $0 < p_s < 1/2$, Algorithm 1 stops no later than*

$$S \leq \min\{m, T + L_n(p_s) - 1\}. \quad (8)$$

Consequently,

$$Q_{\text{gen}}, Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, S), \quad Q_{\text{score}} \leq Q_{\text{mem}}, \quad (9)$$

where $\text{Vol}(m, s) = \sum_{w=0}^{\min\{m, s\}} \binom{m}{w}$.

Proof sketch. The transmitted word is discovered by shell T and contributes at least $n^{-1}p_s^T(1 - p_s)^{m-T}$. Put $s_0 = T + L_n - 1$. If $s_0 \geq m$, finite exhaustion gives $S \leq m$. Otherwise U_{s_0} is defined and its ratio to that component is $n\rho^{L_n} < 1$, so the strict certificate fires by s_0 . Counting $n + 1$ distinct insertion outputs per error vector gives (9). $\square$

If τ_δ is the smallest binomial quantile satisfying $\Pr[T \leq \tau_\delta] \geq 1 - \delta$, then with probability at least $1 - \delta$,

$$Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, \min\{m, \tau_\delta + L_n - 1\}). \quad (10)$$

For fixed p_s and any $c > 0$, Hoeffding concentration and standard Hamming-ball asymptotics give, with probability at least $1 - n^{-c}$,

$$Q_{\text{mem}}, Q_{\text{score}} \leq 2^{n(h_2(p_s) + o(1))}. \quad (11)$$

This exponent mechanism is the classical BSC typical-volume/GRAND mechanism [3], transported to the exact synchronization decoder after Theorem 3; it is not claimed as a new entropy law. Formally, with $a_n = p_s + \sqrt{c \ln n / (2m)} + (L_n - 1)/m$, the high-probability event has $S/m \leq a_n$, $a_n = p_s + o(1)$, and for all sufficiently large n , $a_n < 1/2$. Monotonicity and continuity give $\text{Vol}(m, S) \leq 2^{mh_2(a_n)} = 2^{n(h_2(p_s) + o(1))}$.

V. RANDOM-CODE OCCUPANCY BENCHMARK AND EXACT EVIDENCE

The large reduction in complete scoring can be predicted without attributing it to code structure.

Proposition 4 (Uniform-code occupancy benchmark). *Fix a queried set $A \subseteq \{0, 1\}^n$ of size G containing the transmitted word x_0 . Let $\mathcal{C}$ be a uniformly chosen M -word codebook conditioned on $x_0 \in \mathcal{C}$. Then*

$$\mathbb{E}[|A \cap \mathcal{C}| \mid A, x_0] = 1 + (G - 1) \frac{M - 1}{2^n - 1}. \quad (12)$$

Proof. Each of the other $G - 1$ queried words belongs to the remaining uniformly selected $M - 1$ codewords with probability $(M - 1)/(2^n - 1)$; linearity of expectation proves the result. $\square$

This is a benchmark for a fixed query set, not a theorem for structured codes or for data conditioned on a stopping event. At $n = 32$ and shell one, the generation cap is $G_1 = (n + 1) \text{Vol}(31, 1) = 1056$. Equation (12) predicts mean complete-score counts 1.52, 5.12, and 17.48 for $k = 21, 24, 26$, respectively, close to the observed medians.

The frozen campaign comprised 996 exact trials: 896 natural and 100 forced-weight stress trials, with zero validation violation or censoring. Deduplicated shell search and the history-pair search returned identical ML scores and complete tie sets; 384 exhaustive and 432 completed generic branch-and-bound comparisons also agreed. At $n = 32$, all five cells had median stop shell one. Only 16 exhaustive $n = 32$ timing calibrations were run, all at $k = 21$; their pooled median exhaustive/FIBER time ratio was 465.6. Branch-and-bound was faster in one of five $n = 32$ cells and slower in four, so no universal exact-decoder speed claim follows.

The classical $n + 1$ generator reduced attempted insertions by about $1.94\times$ relative to $2n$ history pairs, but failed its preregistered wall-clock gate. Selected first-shell representatives differed from selected aggregate-ML representatives in many trials, but the tie sets always overlapped; zero sampled case had disjoint tie sets. Theorem 1, not empirical prevalence, supplies the strict pathwise failure. High ML block-error rates in these high-rate cells are

channel/code behavior, not decoder incorrectness, because all exact comparators agree.

VI. CONCLUSION

The exact aggregate-likelihood principle, minimum deletion-alignment metric, membership-oracle interface, and generic incumbent stopping logic are prior art. The defensible one-deletion results are narrower: an explicit strict component/aggregate reversal, the closed-form unseen-shell bound U_s , complete-tie exact certification using that bound, and the realization-dependent stop $S \leq T + L_n - 1$. Standard entropy-volume arguments then yield the usual high-probability exponent. The occupancy benchmark quantitatively explains the finite scoring savings. Together these form a correct, auditable workshop-scale contribution for arbitrary membership-testable binary codes under the stated channel, without implying a new universal synchronization-decoding paradigm.

REPRODUCIBILITY STATEMENT

All numerical values are frozen from immutable commit `2f7736b70bf7f6f659a6707fad716407635a7deb`. The exact C++ source, trial-level outputs, theorem checks, and manifests are available in the public repository https://github.com/afazeliUofT/Grand_Work. Phase 2E performs no new simulation.

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TABLE I

CODE-INDEPENDENT FINITE-CONFIDENCE MEMBERSHIP CAPS FROM (10) AT $p_s = 0.05$. THESE ARE CONSERVATIVE THEOREM BOUNDS, NOT MEASURED WORK.

n	δ	L_n	τ_δ	$(n+1) \text{Vol}(n-1, \tau_\delta + L_n - 1)$
16	10^{-2}	1	3	9,792
16	10^{-3}	1	4	32,997
24	10^{-2}	2	4	1,113,800
24	10^{-3}	2	5	3,637,475
32	10^{-2}	2	5	31,107,417
32	10^{-3}	2	6	117,883,392

TABLE II

FROZEN $n = 32$, $p_s = 0.05$ SCORING EVIDENCE. SAVINGS COUNT AVOIDED COMPLETE LIKELIHOOD EVALUATIONS, NOT UNIVERSAL LATENCY.

Cell	med. Q_{mem}	med. Q_{score}	med. savings
CRC, $k = 21$	970.5	3.0	6.99×10^5
Random, $k = 21$	972.0	1.5	1.57×10^6
CRC, $k = 24$	964.5	4.0	4.19×10^6
Random, $k = 24$	966.5	4.0	4.19×10^6
Ext. Hamming, $k = 26$	954.0	13.5	4.98×10^6

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